

Remote sensing of global greening: Bibliometric evolution, thematic structure, and observation uncertainties

Yufeng Liu^a, Xiaohan Sun^a, Cheng Huang^a, Xinpei Yu, Aming You^a, Shuangshuang Li^b,
Jinxiu Liu^b, Kai Yan^{a,*}

^a State Key Laboratory of Remote Sensing and Digital Earth, Advanced Interdisciplinary Institute of Satellite Applications, Faculty of Geographical Science, Beijing Normal University, Beijing, China, 100875.

^b School of Artificial Intelligence, China University of Geosciences, Beijing 100083, China

Highlights

- Bibliometric mapping traces global greening research from 1980 to 2025.
- Remote sensing studies shifted from greenness detection to attribution.
- Multi-source indicators expose structural-functional greening uncertainty.
- Climate, water, land use and restoration jointly shape greening patterns.
- Future work should link greenness with carbon, water and resilience.

Abstract

Global greening is a major signal of terrestrial ecosystem change, but its research trajectory, methodological basis and ecological meaning remain unevenly synthesized. This review maps remote-sensing-based global greening research from 1980 to 2025 using bibliometric analysis, keyword clustering, co-citation networks and supplementary MODIS-era trend illustrations. Based on 498 Web of Science Core Collection records, we examine publication growth, country and institutional output, disciplinary and journal patterns, method use, thematic evolution and future scenario evidence. The literature expanded slowly from early satellite-based greenness detection and then rapidly after 2010, alongside wider use of

MODIS, Landsat, Sentinel observations, Google Earth Engine, machine learning and ecosystem models. Three linked research fronts now organize the field: climate-change and carbon-cycle mechanisms, multi-source observation and modelling methods, and regionally heterogeneous responses associated with phenology, ecological restoration, agricultural management, water constraints and land degradation. The synthesis shows that global greening is widely reported but cannot be interpreted as uniform ecosystem improvement. Indicator choice, sensor history, spatial scale, masking strategy and attribution method strongly affect estimated greening patterns. Increases in normalized difference vegetation index or leaf area index may also decouple from carbon storage, evapotranspiration, soil moisture and ecosystem resilience. Future work should connect long-term optical records with functional indicators, process models, ground observations and causal attribution frameworks to distinguish structural greening from sustainable ecosystem recovery.

Keywords

Global greening; Remote sensing; Bibliometric analysis; Vegetation indices; Attribution uncertainty; Carbon-water coupling; Ecosystem resilience

1. Introduction

Vegetation is a key part of terrestrial ecosystems, regulating climate, maintaining the carbon cycle, shaping water and heat exchange, and supporting biodiversity (Miralles et al., 2025; Li et al., 2024). Reliable quantification of vegetation status and its changing patterns is essential for explaining global change, evaluating ecosystem functions, and guiding ecological and environmental management. Over the past few decades, climate change has intensified, and human activities have increasingly altered land surface processes. Against this background, terrestrial vegetation has shown a sustained increase across many parts of

the world, and the issue of “global greening” has drawn growing attention in global change research (Myneni et al., 1997; Zhu et al., 2016; Piao et al., 2019). In most studies, global greening refers to long-term increases in vegetation greenness, leaf area index (LAI), net primary productivity (NPP), and related indicators. Beyond simple indicators of biomass accumulation, these dynamics are fundamentally linked to carbon sequestration, ecosystem stability, and regional environmental variability (Zhu et al., 2016; Ding et al., 2020; Zeng et al., 2018).

Multi-source remote sensing and cloud-computing platforms have allowed researchers to examine the characteristics, drivers, and feedbacks of global greening. Observational evidence shows that although greening is a widespread global signal, it shows strong spatial and temporal heterogeneity (Zhang et al., 2003; Radočaj et al., 2020). The research focus has also moved from early-stage phenomenological identification toward mechanistic attribution. Global greening is a coupled process driven by rising atmospheric CO₂ concentrations (fertilization effect), climate change (warming and precipitation shifts), and intensive human interventions such as ecological restoration and agricultural management (Zhu et al., 2016; Piao et al., 2019; Huang et al., 2021). Recent studies also examine the ecological consequences of this greening, particularly its non-linear feedbacks on local evapotranspiration, soil moisture constraints, and land surface temperature regulation (Teng et al., 2023; Jin et al., 2024).

Existing reviews provide a useful starting point, but many focus on a single vegetation index, ecosystem or application context (Xue et al., 2017; Vidican et al., 2023; Wang et al., 2022a; Radočaj et al., 2023). The contribution of this review is to connect three strands that are often treated separately: the bibliometric evolution of the field, the remote-sensing observation systems that support greening detection, and the ecological interpretation of structural greenness. This framing is important because publication growth, indicator choice

and attribution methods have changed together. A field-level synthesis is therefore needed to show how research topics, methods and disciplinary links have co-evolved, and to identify where greening evidence remains observationally or functionally uncertain.

Bibliometric analysis helps address this gap by tracing patterns across authors, institutions, countries, journals, keywords and cited references. Co-occurrence, clustering and temporal analyses can reveal recurring themes and shifts in research attention. For global greening, these results are useful beyond publication counts. They provide preliminary evidence of international participation and possible collaboration patterns, although dedicated co-authorship metrics are still needed to quantify collaboration strength.

This study uses bibliometric analysis, supported by selected MODIS-era remote-sensing trend illustrations, to examine publication trends, country and institutional output, journal and disciplinary patterns, co-citation networks and major thematic evolution in global greening research. It also discusses the research areas that have shaped the field, including climate-vegetation coupling, carbon-cycle feedbacks, remote-sensing observation, model simulation, phenology, ecological restoration and regional applications. The MODIS-era NDVI, LAI and GPP trend maps and global area-weighted annual means are used to connect literature patterns with representative observations. They are not presented as a full causal attribution of global greening.

The analysis is organized around publication and contribution patterns, thematic clusters and technical support systems, and the main uncertainties in observation and attribution. The central argument is that global greening should be evaluated as a coupled measurement and ecosystem-function question, rather than as a simple increase in greenness values.

2. Current status of global greening research

This section reviews the development of global greening research since the 1980s. It first traces how the field moved from remote-sensing observation and indicator characterization toward mechanism interpretation and integrated assessment. It then uses bibliometric evidence to examine publication trends, country and institutional patterns, and disciplinary and journal structures.

2.1. Publication output trends

Publication output partly reflects the development of research interest in global greening. Fig. 1 shows annual publication trends across spatial scales. Since the 1980s, studies have moved from initial phenomenon identification and regional verification toward global-scale analysis and questions about underlying drivers (Mouat et al., 1993; James et al., 1994). In the early period, remote sensing technology was still limited and long time-series data were not widely available. Researchers therefore relied mainly on coarse-resolution datasets such as AVHRR to identify vegetation change at local or regional scales (James et al., 1994; Reed et al., 1994; Coppin et al., 1996). This stage provided the basis for later work, although the available evidence was still relatively coarse.

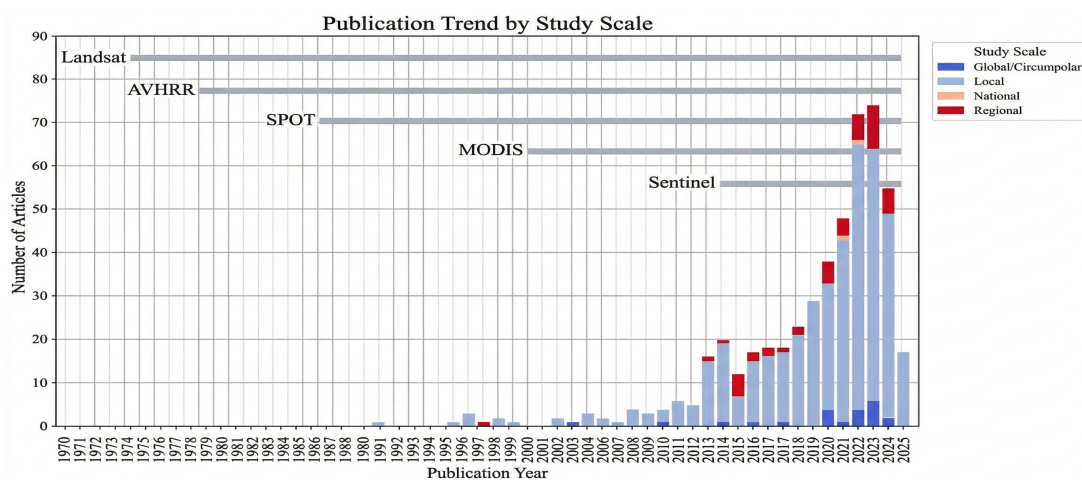


Fig. 1. Annual publication output in global greening research across spatial scales (1980s-2025) and timeline of major satellite sensor applications.

The situation changed after the beginning of the 21st century. With the wide use of remote sensing products such as Moderate Resolution Imaging Spectroradiometer (MODIS), which offered medium and high temporal resolution, the ability to monitor vegetation dynamics at the global scale improved greatly (Zhang et al., 2003; Beck et al., 2005; De Jong et al., 2010). What had once appeared as scattered observations from individual regions gradually came to be understood as a global signal linked to climate change. As long-term remote sensing records accumulated, more and more studies reported that a substantial share of terrestrial vegetation showed persistent greening trends. This expanding body of evidence has helped shift the field from identifying where greening occurred to explaining its underlying drivers (Myneni et al., 1997; Zhou et al., 2001; Fensholt et al., 2012).

After the early 2010s, global greening research entered a period of rapid expansion. Landsat and MODIS applications continued to mature, and the increasing use of Sentinel observations after the mid-2010s improved the capacity to characterize spatial and temporal heterogeneity in vegetation change (Fensholt et al., 2012; Clark et al., 2010; Radočaj et al., 2020). New research tools also changed how the problem could be studied. Platforms such as Google Earth Engine, together with machine learning methods and Earth system models, gave researchers stronger ways to evaluate the relative roles of CO₂ fertilization, climate change, human activities, and ecological engineering in shaping greening patterns (Fensholt et al., 2012; Scharlemann et al., 2008). As methods improved, the research questions broadened. Researchers no longer asked only whether greening was happening; they also asked why it was happening, whether it meant ecological improvement, and whether it could continue under future environmental conditions (De Jong et al., 2010; Fensholt et al., 2012; Beck et al., 2005).

The literature nevertheless shows that global greening is neither simple nor spatially uniform. Most studies agree that greening is widespread, but they also identify strong

regional differences and complex driving mechanisms (Zhu et al., 2016; Piao et al., 2019; Zhao et al., 2017). Rising atmospheric CO₂, warming, precipitation shifts and ecological restoration projects are all widely recognized as important drivers (Zhu et al., 2016; Zhao et al., 2017; Huang et al., 2021). Their relative importance varies among regions, ecosystems and time periods (Huang et al., 2021; Teng et al., 2023; Jin et al., 2024). Global greening therefore cannot be interpreted only as an increase in greenness values. It also requires attention to uncertainty, indicator behavior and ecological meaning. For example, NDVI saturation, sensor-transition effects and the question of whether higher greenness reflects greater biomass or productivity all affect interpretation (Guay et al., 2014; Qiu et al., 2021; Wang et al., 2022b). Differences among conclusions often arise not only from ecological variation, but also from indicator choice, data source and processing workflow. The causes, consequences and long-term sustainability of global greening therefore remain open to debate.

2.2. Country and institutional patterns

Academic output in the field of global greening research shows a clear pattern of national concentration (Fig. 2). China leads with 167 publications; the United States (U.S.) ranks second with 153 publications. They are followed by Germany (31), Canada (29), France (25), Australia and the United Kingdom (24 each), India and Sweden (14 each), and Switzerland (12). The distribution is concentrated mainly in the Northern Hemisphere, with China and the United States holding a clear lead in publication output. This pattern likely reflects long-term remote-sensing capacity, national ecological-restoration programs, climate-change research demand, and sustained research investment. Countries such as the United Kingdom, Germany, France, and Canada also remain active contributors, while Australia and India add case-study contexts with ecologically fragile or environmentally sensitive regions.

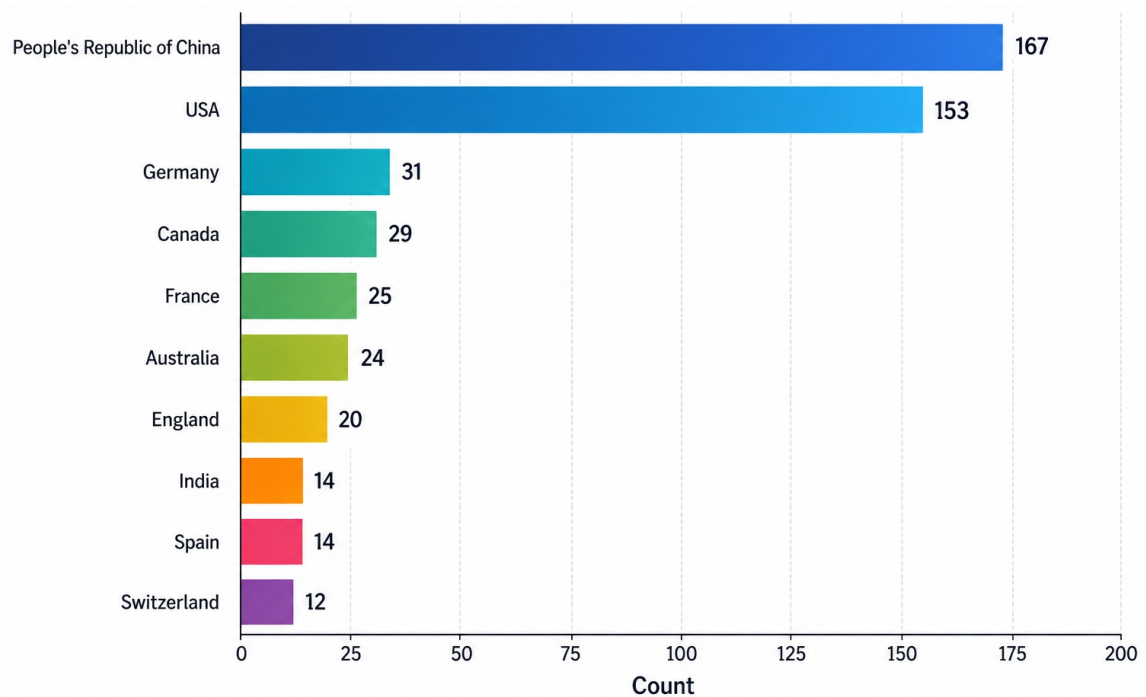


Fig. 2. Major contributing countries in global greening research and their publication frequencies.

Institutional output shows a similarly concentrated pattern (Fig. 3). The Chinese Academy of Sciences published the largest number of papers (123), followed by Beijing Normal University (77), the University of Chinese Academy of Sciences (73) and Peking University (71). Lanzhou University and Nanjing University published 53 and 54 papers, respectively; Sun Yat-sen University and Tsinghua University each published 52 papers; and Wuhan University published 43 papers. Among U.S. institutions, the National Aeronautics and Space Administration (NASA) ranked first with 49 papers. Boston University, Columbia University, the Jet Propulsion Laboratory, Northern Arizona University, the University of Arizona and the University of California, Berkeley each published 41 papers. Overall, academic output from Chinese institutions is more concentrated, whereas U.S. output is more distributed across multiple centers. Because project-level collaboration information was not available, these findings are based only on institutional affiliations and should be regarded as indicative, not definitive.

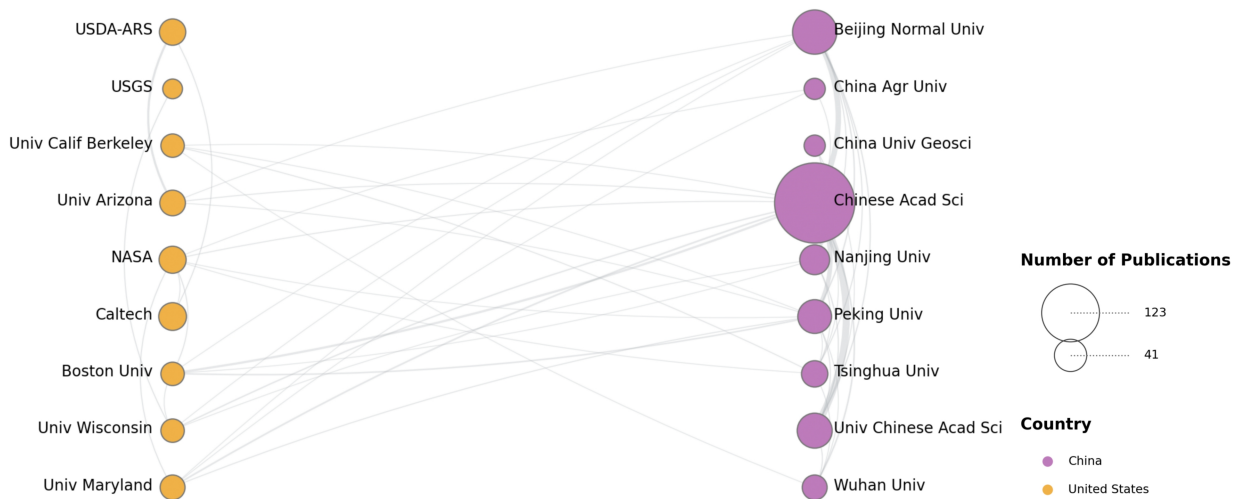


Fig. 3. Core publishing institutions in global greening research.

2.3. Disciplinary and journal distribution

By discipline, research on global greening is highly interdisciplinary. Environmental science is the primary focus, followed by remote sensing, photogrammetry, geographic information systems and ecology. The field relies on remote-sensing observations and data-processing methods, but it is also closely tied to climate change, biogeochemical cycles and ecosystem assessment. Machine learning, geoscience big data, ecological modelling and sustainable-development studies now appear more often, indicating a gradual shift from monitoring toward mechanism analysis and applied assessment.

Journal distribution shows the same interdisciplinary pattern. As shown in Table 1, Remote Sensing, Remote Sensing of Environment, Environmental Research Letters, Geophysical Research Letters and International Journal of Applied Earth Observation and Geoinformation are among the main publication outlets. Remote Sensing has the largest publication volume, reflecting its strong coverage of methodological applications and case studies. Remote Sensing of Environment has high citation impact and remains central to theoretical, methodological and broad-scale remote-sensing research. The publication pattern

is therefore centered on remote-sensing journals and extends outward to ecology and climate-science journals.

Table 1

Top 15 journals publishing global greening research, ranked by publication count.

No.	Journal	Publications	CiteScore	Impact Factor (IF)	H-index
1	Remote Sensing	57	8.60	4.1	81
2	Agricultural and Forest Meteorology	27	11.10	5.7	144
3	Science of the Total Environment	27	16.40	8.0	205
4	Environmental Research Letters	24	11.90	5.6	97
5	Global Change Biology	23	21.50	12.0	217
6	Remote Sensing of Environment	18	22.60	11.4	238
7	Ecological Indicators	16	11.80	7.0	97
8	Geophysical Research Letters	16	9.00	4.6	337
9	Sustainability	16	6.80	3.3	207
10	Journal of Geophysical Research- Biogeosciences	10	6.10	3.5	73
11	International Journal of Applied Earth Observation and Geoinformation	10	13.50	8.6	76
12	Land	10	5.90	3.2	71
13	Biogeosciences	8	8.60	3.9	106
14	Journal of Cleaner Production	8	15.80	10.0	268
15	Frontiers in Plant Science	8	8.80	4.8	83

*CiteScore, Journal Impact Factor, and h-index were collected from Scopus, Journal Citation Reports, and Google Scholar Metrics, respectively

2.4. Co-citation analysis

Co-citation analysis is useful for identifying the knowledge base, core literature, and evolutionary trajectory of a research field. Overall (Fig. 4), research on global greening has formed a clear co-citation knowledge network.

Among the core publications (Table 2), Huete et al. and Tucker et al. mark the early methodological base for vegetation indices and remote-sensing greenness monitoring. Myneni, Nemani, and related studies helped turn these methods into evidence for large-scale

vegetation change, while Zhu, Chen, and Piao shifted the discussion toward drivers, regional contributions, carbon-cycle links, and global-change assessment. Gorelick et al. also appears as an important node because Google Earth Engine changed the computational scale at which long-term vegetation records could be processed. The co-citation network therefore shows a knowledge path from index development and greening detection to attribution, regional comparison, and ecosystem feedbacks.

Table 2

Highly co-cited publications in global greening research.

Title	Journal	First author	Year	Co-citation frequency
Overview of the radiometric and biophysical performance of the MODIS vegetation indices	Remote Sensing of Environment	Huete a	2002	89
Greening of the Earth and its drivers	Nature Climate Change	Zhu zc	2016	66
Red and photographic infrared linear combinations for monitoring vegetation	Remote Sensing of Environment	Tucker cj	1979	60
China and India lead in greening of the world through land-use management	Nature Sustainability	Chen c	2019	56
Increased plant growth in the northern high latitudes from 1981 to 1991	Nature	Myneni rb	1997	50
Climate-Driven Increases in Global Terrestrial Net Primary Production from 1982 to 1999	Science	Nemani rr	2003	50
Characteristics, drivers and feedbacks of global greening	Nature Reviews Earth & Environment	Piao sl	2020	46
An extended AVHRR 8-km NDVI dataset compatible with MODIS and SPOT vegetation NDVI data	International Journal of Remote Sensing	Tucker cj	2005	36
Google Earth Engine: Planetary-scale geospatial analysis for everyone	Remote Sensing of Environment	Gorelick n	2017	30
Greenness in semi-arid areas across the globe 1981–2007 — an Earth Observing Satellite based analysis of trends and drivers.	Remote Sensing of Environment	Fensholt r	2012	29

The co-cited literature can be grouped into three broad strands: remote-sensing methods, climate-ecology coupling, and global-change synthesis. Remote-sensing papers provide the

basis for identifying and monitoring greening, while climate and ecology papers connect greening to drivers, attribution, and ecological feedbacks. Papers in Science, Nature, and related journals have also increased the visibility of the field. The co-citation network points to a clear shift from monitoring greening to explaining and assessing it.

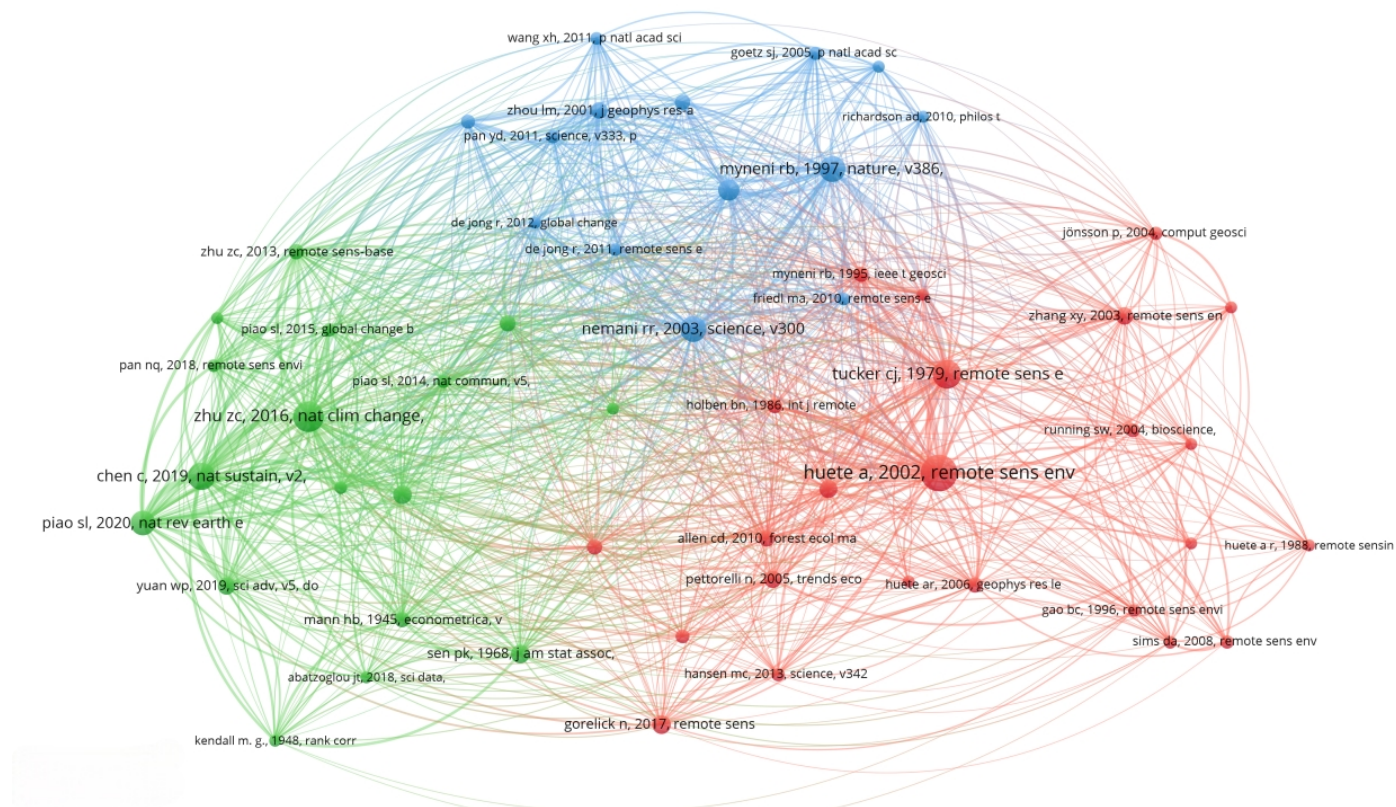


Fig. 4. Co-citation network of studies on global greening.

2.5. Thematic evolution

As condensed markers of subject matter and disciplinary focus, keywords help capture the core issues, knowledge structure and evolutionary trends of global greening research (Reed et al., 1994). Keyword analysis, particularly the identification of frequent terms and co-occurrence patterns, is an important bibliometric method for tracking research hotspots and frontier shifts over time (Ichii et al., 2001).

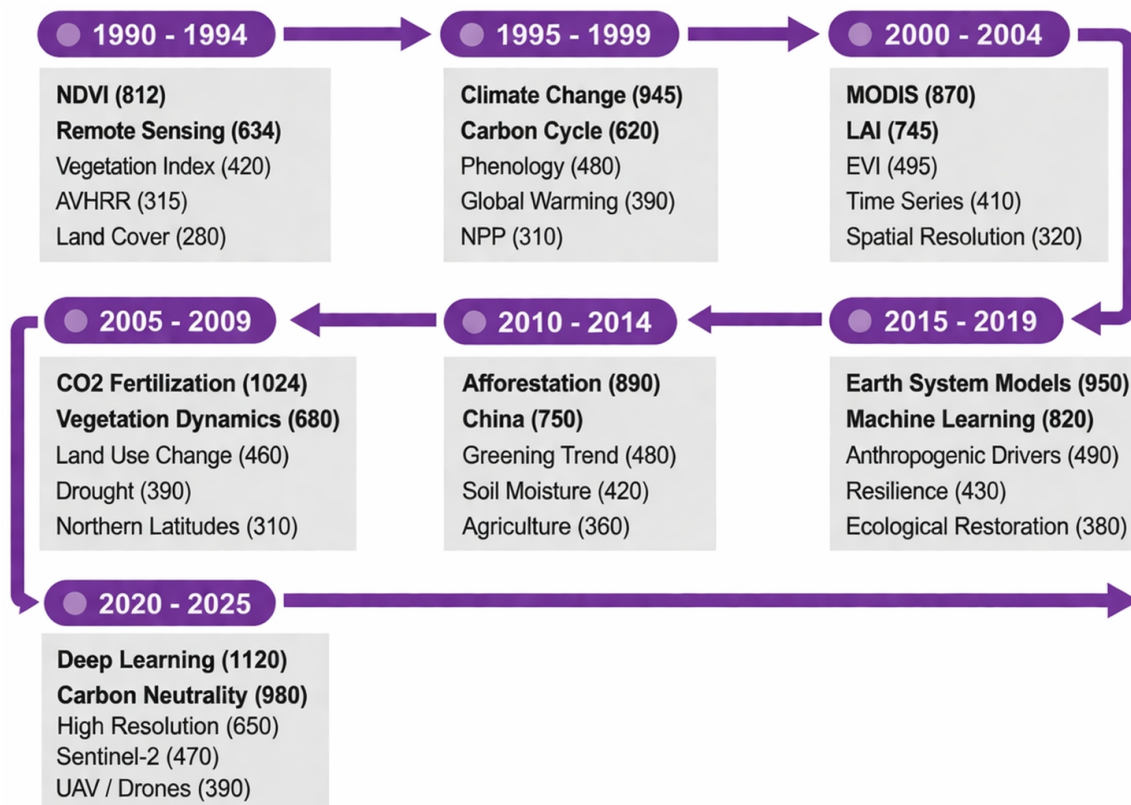


Fig. 5. Temporal evolution of high-frequency keywords in global greening research, 1990–2025

Fig. 5 illustrates the temporal distribution of high-frequency keywords in global greening research from 1990 to 2025. The field has generally progressed from remote-sensing observation and greenness identification to climate-carbon mechanism analysis, and then to model prediction and intelligent analysis. In the early stages, research centered on keywords such as NDVI, remote-sensing monitoring, vegetation indices, AVHRR and land cover. This indicates that the main task was to identify vegetation-change phenomena from remote-sensing observations and establish a foundational data framework for global greening research (Reed et al., 1994; Loveland et al., 2000). Later, keywords such as climate change, carbon cycle, phenology, global warming and net primary productivity became more prominent, indicating a shift from phenomenon identification toward mechanisms and ecological or climatic effects (Nemani et al., 2003; Lucht et al., 2002).

Since the beginning of the 21st century, keywords such as MODIS, LAI, enhanced vegetation index (EVI), time series, and spatial resolution show that research has increasingly

focused on long-term monitoring and scale effects supported by multi-source remote sensing data (Zhang et al., 2004; Yang et al., 2006; Scharlemann et al., 2008). After 2005, keywords such as CO₂ fertilization, vegetation dynamics, land-use change, drought, and high-latitude regions became more common, reflecting growing attention to driver attribution and regional variation (Piao et al., 2006; Scheiter et al., 2008; Notaro et al., 2005). After 2010, afforestation, China, greening trend, soil moisture, and agriculture appeared more often, showing a shift from global pattern description toward key regions and ecological processes, especially the roles of human activities, ecological engineering, and resource constraints (Piao et al., 2014; Su et al., 2018; Zhang et al., 2015). In recent years, terms such as “Earth system models”, “machine learning”, “anthropogenic drivers”, “ecological restoration”, “deep learning”, “carbon neutrality”, high resolution, Sentinel-2, and Unmanned Aerial Vehicle (UAV) have also appeared more often. Rather than marking a fully integrated new phase, these terms show that recent studies increasingly combine multi-source remote sensing, Earth system models, and machine-learning approaches, although integration remains uneven across regions and methods. Research objectives have also expanded to include ecological restoration assessments, terrestrial ecological feedbacks under carbon neutrality, and future scenario projections (Brandt et al., 2018; Tong et al., 2016).

2.6. Research methods

The methodological framework of global greening research did not begin with the layered structure seen today. Research approaches have moved from single-trend identification toward multi-method analysis: early studies focused on whether greenness had changed and whether the change was significant, while more recent studies examine quantitative relationships between greenness and factors such as temperature, precipitation, soil moisture, and human activities. Among the high-frequency methods identified in this study, correlation analysis appeared in 71 papers, the Mann–Kendall test in 63, trend analysis

in 59, linear regression/OLS in 54, index construction/index comparison in 53, Sen's slope/Theil–Sen in 50, partial correlation analysis in 30, and random forests in 21. The field still relies heavily on statistical analysis and trend tests, but nonlinear modeling, process simulation, and attribution analysis are becoming more visible. These method frequencies were obtained from method-related keyword extraction and manual screening; because one paper may use multiple methods, the counts are non-exclusive.

Among the method clusters, trend identification and change detection are foundational. They account for 97 publications, the largest and most frequently cited category. Researchers usually use the Mann–Kendall test, Sen's slope/Theil–Sen, general trend analysis, and time-series analysis to test whether greenness has changed, how fast it has changed, and whether the temporal process has distinct phases ([Wang et al., 2020](#)). Breakpoint detection (BFAST), the Hurst index, RESTREND, Logistic/Gompertz fitting, GAM, bootstrap resampling, and standardized anomaly analysis are used to identify abrupt shifts, assess persistence, or detect anomalous events ([Sharma et al., 2016](#)). Once a study establishes that greenness has changed, the next question is usually what drives the change ([He et al., 2023](#)). Regression models are widely used for this purpose, with 72 relevant studies, including 54 that used linear regression/OLS. Other methods include multivariate regression, autoregressive and panel regression, GLS/GLM/GLMM/LMM, GAM, ridge regression, GWR/GTWR, and structural equation modeling ([Pourmohammadi et al., 2021](#)). Compared with univariate analysis, regression models can include multiple drivers and estimate the direction, strength, and significance of each variable. Geographically weighted regression and spatiotemporally weighted regression add spatial heterogeneity to the analysis, so the relationship between greenness and its drivers is treated as a regionally varying response rather than a homogeneous process ([Lu et al., 2024](#)).

Alongside regression methods, correlation analysis is also frequently used, with a total of 71 studies employing this approach, including 30 that used partial correlation analysis. Correlation analysis has long been widely used because it provides a direct and intuitive statistical approach for examining associations between vegetation greenness and potential environmental or anthropogenic factors (Algorta et al., 2020). Pearson or Spearman correlation analysis is typically used to evaluate univariate relationships between greenness and variables such as temperature, precipitation, evapotranspiration, and soil moisture. Partial correlation analysis, on the other hand, can be used to examine the association between greenness and a specific factor after statistically controlling for selected covariates; therefore, it is often adopted as an exploratory tool when comparing the relative roles of climatic and anthropogenic influences. However, such conditional associations should not be interpreted as direct causal effects without further support from process-based models, long-term observations, field experiments, counterfactual simulations, or multiple independent datasets. Methodological discussions on correlation and partial correlation also remind us that correlation coefficients are sensitive to distributional assumptions, sample structure, variable selection, and the interpretation of control variables (De Winter et al., 2016).

Methods such as PCA, ANOVA, and model comparison do not identify drivers directly, but they help with dimensionality reduction, factor screening, and tests of between-group differences. Indicator construction and comparison are also common, with 53 papers, a number close to those for linear regression and Sen's slope. Researchers use index construction and comparison, diversity or similarity indices, standardization, outlier handling, classification, accuracy assessment, and observation-response analysis to make complex greening processes comparable. These approaches are useful for comparing greenness indices, distinguishing greening intensity across ecological regions or land-use types, and building

evaluations of climate sensitivity, human disturbance, or ecological restoration. Their main role is comparison and classification, not causal attribution.

Machine learning and explainable AI have become more common as studies move toward multivariate, nonlinear, and high-dimensional problems (Rudin et al., 2022). These methods account for 27 papers in this study, fewer than trend detection, regression, or correlation, but they are now a visible part of method development. Random forests are the most common tool, appearing in 21 papers. Other approaches include ensemble learning, deep learning and neural networks, SHAP, classification methods, sensitivity or threshold analysis, and partial dependence analysis (Murdoch et al., 2019). Compared with traditional regression, machine learning is better suited to complex interactions among temperature, precipitation, soil moisture, land use, and human activity (Samek et al., 2021). Process models and scenario simulations have expanded alongside these data-driven approaches and also account for 27 papers. They include climate and ecological process models, ecohydrological models, one-factor-at-a-time analysis, time-series simulation, curve fitting, and uncertainty analysis. These methods are used to simulate vegetation change under different climate, moisture, carbon-cycle, or ecological-engineering scenarios (Holt et al., 2025). Unlike correlation or regression, process models try to place statistical relationships within physiological and ecological mechanisms. They are therefore useful for questions about why a region continues to green, whether greening increases evapotranspiration or reduces soil moisture, and whether the trend can persist under future climate scenarios (Littell et al., 2011).

In contrast, fewer studies focus on attribution and contribution decomposition; this paper identified only 24 such studies. Even so, they address the most central and challenging question in global greening research: exactly how much each of climate change and human activities contributes (Ribes et al., 2016). To achieve this objective, existing studies have

employed methods such as contribution/attribution/decomposition/relative importance analysis, RESTREND, Geodetector, partial correlation analysis, structural equation modeling, random forests, SHAP, OFAT, multiple regression, panel regression, and generalized linear mixed models (Liu et al., 2022). Such studies do not merely seek to determine whether a relationship exists between variables; rather, they aim to identify the dominant factors and their relative contributions in scenarios where multiple potential drivers coexist and are mutually coupled (Zhang et al., 2023). Temperature, precipitation, radiation, CO₂ fertilization effects, soil background, and ecological engineering are often not independent of one another; therefore, the challenge of attribution analysis lies not in identifying the variables, but in effectively disentangling the intertwined signals (Qian et al., 2021).

The method distribution has a practical order. Trend analysis asks whether greenness has changed; correlation and regression examine which factors are associated with those changes; machine learning and process models handle nonlinear or interacting relationships; attribution and contribution analyses try to separate the relative roles of different drivers. In the broader research context (Fig. 8), this sequence reflects a move from identifying facts to testing relationships and, more cautiously, to examining mechanisms. Traditional statistical methods such as correlation analysis, the Mann–Kendall test, trend analysis, and linear regression still dominate. Random forests, process simulation, and attribution analysis are now used more often, but they have not solved the main methodological problems. Studies still use different greenness indicators, which limits comparability. Many analyses remain at the level of correlation or statistical attribution, and complex models can improve fit while reducing interpretability. Further progress depends on better linking multi-source remote sensing with ground observations, checking statistical results against process-based models, and using machine learning in ways that remain ecologically interpretable.

3. Research themes in global greening studies

This section identifies major keyword clusters in global greening research (Fig. 6) and interprets them in relation to data sources, geographic regions, ecological functional zones, applications and supporting technologies (Fig. 7). The aim is to synthesize the main research themes rather than to list every keyword.

3.1. Cluster 1: Remote sensing observations and data-source characterization

In global greening research, remote sensing data serve as the primary source of information for identifying vegetation changes, characterizing spatiotemporal patterns, and conducting long-term monitoring. In terms of keyword frequency, NDVI appeared 122 times, making it the most frequently cited indicator in this cluster. Greenness monitoring based on vegetation indices therefore remains the dominant technical approach in this literature. Because global greening is a macro-level phenomenon with long temporal evolution and broad spatial coverage, data-source selection is driven mainly by the need for consistency and global extent. MODIS (90 mentions) is the primary data source, followed by Landsat (48 times) and AVHRR (40 times). Higher-resolution sensors such as Sentinel-2 (10 times) are increasingly used for localized analysis, while Sentinel-1 and WorldView-2 each appeared only once. The pattern shows that global greening research still relies mainly on long-term multispectral records with medium spatial resolution; higher-resolution or specialized sensors are used mostly for supplementary regional analysis.

MODIS has become the most widely used data source in global greening research because it provides standardized, high-frequency, and multi-parameter vegetation products. MODIS vegetation index products provide globally consistent 16-day composite observations that are widely used for monitoring vegetation dynamics (Didan et al., 2015). MODIS LAI products further support canopy-structure analysis and have been extensively evaluated and



Fig. 7. Framework of core themes in global greening research: remote sensing observations, temperature effects, regional regulation and water constraints.

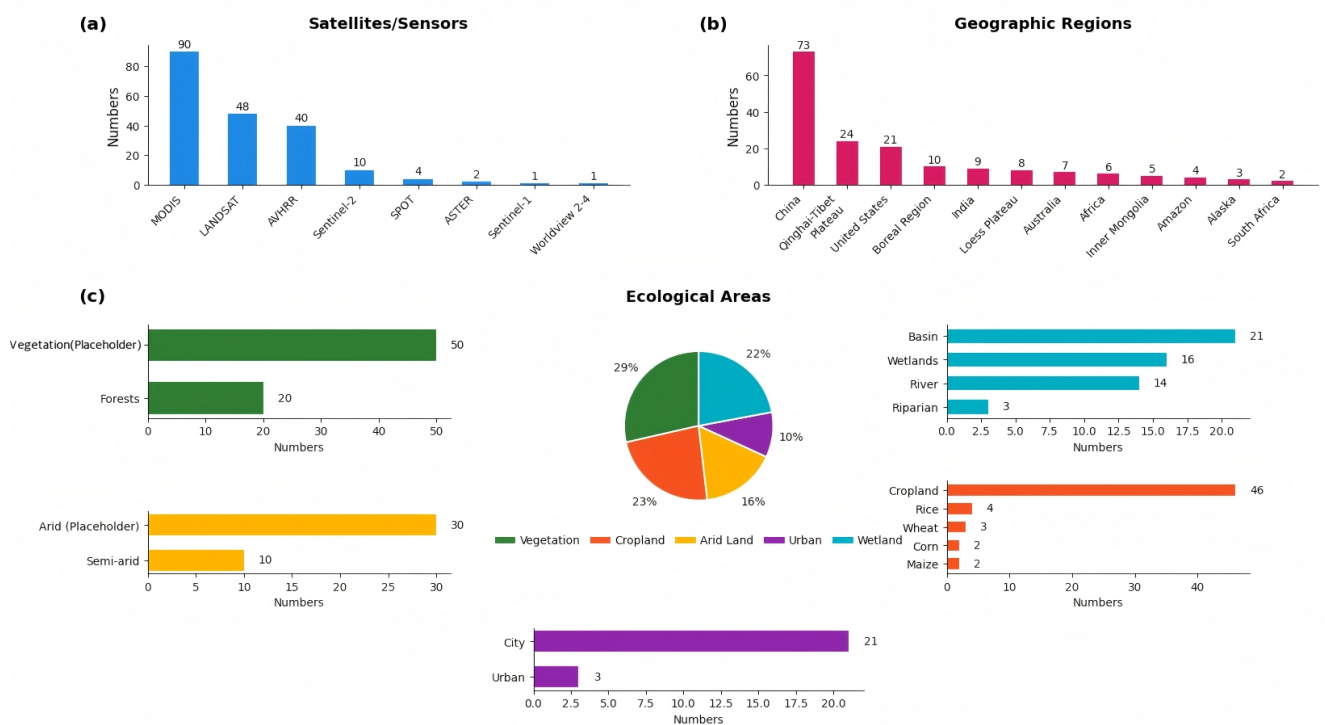


Fig. 8. Multidimensional characteristics of core themes in global greening research: (a) major remote sensing satellites and sensors, (b) representative geographic regions, (c) ecological regions and (d) principal application areas

Landsat provides a complementary perspective because of its much higher spatial resolution and long historical archive. Landsat 8 has strengthened terrestrial global change

observations through improved sensor design and data continuity (Roy et al., 2014). The broader Landsat archive has also become a key resource for long-term land monitoring and land-cover change analysis because of its multi-decadal record and medium spatial resolution (Wulder et al., 2019). More recently, the scientific value of the Landsat program has been further reinforced by its 50-year observational record, which supports retrospective analyses of land surface change and human land-use impacts (Wulder et al., 2022). Although Landsat is less frequently used than MODIS in global-scale greening assessments, it has clear advantages for regional studies, land-use attribution, and the identification of fine-scale vegetation changes. Its 30-m resolution allows researchers to distinguish cropland expansion, afforestation, urban greening, forest disturbance, and land-cover conversion more effectively than coarse-resolution products. Therefore, Landsat is particularly useful for explaining why greening occurs in specific regions and for linking greenness change with human activities or land management. Still, global-scale Landsat-based greening analysis remains computationally demanding and is affected by cloud contamination, data gaps, and the need for cross-sensor harmonization across Landsat missions.

Sentinel-2 has become increasingly important in recent years because of its high spatial resolution, frequent revisit cycle, and red-edge bands. As an optical high-resolution mission, Sentinel-2 was designed to support systematic land monitoring at fine spatial scales (Drusch et al., 2012). Its spectral configuration, especially the inclusion of red-edge bands, improves the estimation of vegetation biophysical variables such as LAI and chlorophyll-related canopy properties (Frampton et al., 2013). These characteristics make Sentinel-2 particularly useful for land-cover and vegetation monitoring applications, especially at local and regional scales (Phiri et al., 2020). However, Sentinel-2A was not launched until 2015 and Sentinel-2B in 2017, so its time series is still too short to independently support long-term global greening trend detection. Its value in global greening research therefore lies mainly in local-

to regional-scale monitoring, validation of coarser products, and the refinement of mechanism studies rather than in reconstructing multi-decadal global greening trajectories.

Recent studies have moved beyond traditional greenness indices toward indicators that separate structural greening from functional ecosystem change. LAI describes canopy structure and leaf area, so it adds information beyond surface greenness. SIF is more directly related to photosynthetic activity and helps test whether greening is accompanied by stronger vegetation function. VOD is sensitive to vegetation water content and aboveground biomass structure, which optical greenness indices do not fully capture. NIRv links near-infrared reflectance with greenness and is increasingly used as a proxy for photosynthetic capacity and productivity. This shift matters because higher NDVI or LAI does not always mean greater carbon sequestration, resilience, or ecological quality. Multi-source indicators are therefore needed to move global greening research from detecting more green vegetation to asking whether ecosystems are functioning better.

Microwave and SAR observations also provide useful complementary information, but they should not be directly equated with optical greenness. AMSR-E, as a passive microwave sensor, can provide information related to soil moisture, snow, ice, water vapor, and precipitation, which helps interpret the hydroclimatic background of vegetation change (Njoku et al., 2003). Multi-sensor land data assimilation further shows the value of combining microwave and other Earth observation products for improving global soil moisture and snow estimates (Zhao et al., 2018). Sentinel-1 SAR data provide all-weather and all-season observation capability, making them useful for characterizing surface moisture, vegetation structure, and disturbance-related processes (Fan et al., 2025). However, because SAR and microwave signals do not directly measure canopy greenness, they are better used as auxiliary datasets for understanding water constraints, biomass structure, and disturbance mechanisms rather than as direct substitutes for NDVI or LAI.

The observational basis of global greening research has moved from coarse, long-term optical records toward multi-source, multi-scale and function-oriented observations. AVHRR/GIMMS remains useful for multi-decadal greening analyses since the 1980s. MODIS provides the most widely used standardized products for the past two decades. Landsat supports regional mechanism analysis and land-use attribution, whereas Sentinel-2 improves fine-scale monitoring and validation. Solar-induced chlorophyll fluorescence (SIF), vegetation optical depth (VOD), near-infrared reflectance of vegetation (NIRv), microwave and SAR data increasingly connect structural greenness with photosynthesis, biomass, water stress and ecosystem function. Future work should rely less on a single greenness indicator and combine long-term consistency, spatial detail and functional information.

3.2. Cluster 2: Temperature effects and ecological process feedbacks

Keywords such as “temperature”, “model”, “carbon”, “evapotranspiration”, “time-series”, and “patterns” show a shift from describing greening phenomena toward examining mechanisms. “Temperature” appears 27 times, “model” 21 times, “carbon” 16 times, and “patterns” 7 times. The direct keyword frequency of “evapotranspiration” and “time-series” is low, with two occurrences each, so these terms should be read as links to mechanism-oriented literature rather than as high-frequency core keywords. Temperature remains the main term in this cluster, and it connects greening research with phenology, carbon exchange, model simulation, and land-atmosphere feedbacks ([Sun et al., 2022](#)).

Time-series analysis has shown that rising spring temperatures have advanced the start of the growing season (SOS) across many Northern Hemisphere ecosystems, while delayed autumn senescence (EOS) may further extend the potential carbon uptake period. Researchers also highlight that the temperature sensitivity of phenology may be nonlinear, because chilling requirements and photoperiod constraints interact with forced warming and

complicate future projections of greening trends (Piao et al., 2019). Where such seasonal extension occurs, it can contribute to increases in annual integrated NDVI or LAI observed by satellites.

Ecological scenario statistics show where these temperature- and feedback-related questions are most often studied. Vegetation ecosystems account for the highest proportion, reaching 29% in ecological-region statistics, with 50 studies on general vegetation and 20 on forests. Wetland ecosystems account for 22%, with “watershed”, “wetland”, and “river” appearing 21, 16, and 14 times, respectively. These figures do not mean that all wetland or watershed studies belong only to the temperature cluster. They show that temperature-related greening questions are often examined together with hydrological and carbon-cycle processes in ecosystems with high vegetation cover or strong climate sensitivity (Zhao et al., 2022; Wu et al., 2010). Although the direct keyword frequency of evapotranspiration is relatively low, related studies and mechanism-oriented literature show that evapotranspiration is an important process linking greening with land-atmosphere feedbacks. Increased leaf area can raise latent heat flux through transpiration and cool the land surface, especially in tropical and temperate zones, although this cooling effect may saturate as vegetation density increases (Zeng et al., 2018).

3.3. Cluster 3: Regional case studies and human-activity management

Keyword analysis was used to identify study areas with clear geographic references. Region selection in global greening studies is closely related to ecological sensitivity, human activity, and ecological governance. This cluster is built mainly around keywords such as “China”, “Loess Plateau”, and “afforestation”; “China” appears 30 times, “afforestation” 3 times, and “Loess Plateau” 1 time. In the regional statistics, China-related research appears 73 times, making it one of the most frequent regional contexts in the

literature. It is followed by the Qinghai-Tibet Plateau (24), the United States (21), northern regions (10), India (9), the Loess Plateau and Australia (8 each), Africa (6), and Inner Mongolia (5), while the Amazon, Alaska, and South Africa appear 4, 3, and 2 times. These patterns show that global greening studies concentrate on fragile ecosystems, climatic transition zones, and regions strongly affected by human activity.

China is one of the most frequently examined regional contexts in this literature. This is related to its diverse eco-geographic patterns as well as large-scale ecological restoration projects and land-use transformations in recent years (Song et al., 2022). The Qinghai-Tibet Plateau, often referred to as the “Roof of the World”, is a key region for studying vegetation dynamics and ecological responses because of its high altitude, low temperatures, and sensitivity to climate change (Liu et al., 2025). The Loess Plateau is a typical region of soil erosion and ecological restoration in northern China. Against the backdrop of the Grain-for-Green Program and large-scale afforestation, the relationship among vegetation restoration, soil moisture consumption, and regional hydrological responses has attracted sustained attention (Fan et al., 2024). Inner Mongolia, located within the Eurasian steppe zone, is a major pastoral and grassland ecological region in China. Changes in grassland resources, grazing intensity, and climate variability make it a useful case-study area for global greening research (Miao et al., 2021).

Beyond China, the United States is also a key regional context in this field (Yu et al., 2018). The agricultural belt and arid and semi-arid regions of the central United States, shaped by intensive agricultural management, irrigation and land-cover change, are frequently used to examine how human activities regulate vegetation greenness and ecological processes (Gurgel et al., 2025). High-latitude northern regions, especially coniferous forest and tundra transition zones, are important for studying temperature-driven vegetation change under warming (Parra et al., 2024). The Amazon is a key region for

examining tropical carbon sinks, forest disturbance and climate feedbacks (Nunes et al., 2022). Arid and semi-arid regions in Africa, Australia and South Africa more clearly show the combined effects of water constraints, desertification risk and human land-use management (Olorunfemi et al., 2021).

From an ecological-region perspective, agricultural ecosystems account for 23% of the total, including 46 studies on agricultural ecosystems in a broad sense and smaller sets focused on paddy fields, wheat, corn and maize. Urban ecosystems account for 10%, with 21 studies. These patterns indicate that global greening research increasingly examines not only natural ecosystem responses to climate change, but also the regulatory effects of agricultural management, afforestation, urban expansion and land-use transitions. In China's Loess Plateau, for example, cropland conversion and large-scale afforestation have increased vegetation cover but also raised concerns about soil moisture depletion, reduced runoff and restoration sustainability (Yu et al., 2020). In Inner Mongolia and the agro-pastoral transition zone of northern China, grazing intensity, grassland management and ecological restoration policies jointly shape vegetation-greenness patterns (Liu et al., 2025). The U.S. Midwest illustrates how irrigation, cropping-system adjustments and intensive land management can rapidly alter vegetation cover and growing-season processes (Yu et al., 2018). Urban green-space development, ecological restoration and landscape reconstruction have also made urban areas important settings for studying anthropogenic regulation of vegetation change (Xing et al., 2023). China, the Qinghai-Tibet Plateau, the Loess Plateau, Inner Mongolia, the U.S. agricultural zone, and the Amazon have become important regions for studying spatial heterogeneity in global greening and the effects of human activity.

3.4. Cluster 4: Water constraints and regional ecological vulnerability

Keywords such as drought, water, forest variability, semiarid ecosystems and river basin show the importance of moisture constraints in global greening research. This cluster focuses mainly on arid, semi-arid and water-sensitive regions. Arid ecosystems account for 16% of all ecological-region records, including 30 mentions of broadly defined arid zones and 10 mentions of semi-arid zones. Wetland and river-related regions together account for 22%, indicating that basin-scale hydrology has become an important lens for explaining heterogeneity in vegetation change. The Qinghai-Tibet Plateau, northern China, Africa, the Amazon and South Africa all receive repeated attention in this cluster.

Arid and semi-arid regions are typical areas for studying the effects of water constraints on vegetation change (Li et al., 2021). For example, the Sahel region of Africa is often used to analyze the relationship between precipitation recovery, improved soil moisture, and regional “re-greening”, making it one of the most representative case studies in global research on vegetation change in arid regions (Ye et al., 2018). In China’s northwestern arid and semi-arid regions, particularly Xinjiang, the Hexi Corridor, and the Inner Mongolia grassland transition zone, research has focused more on exploring vegetation dynamics under the combined influence of climate variability, irrigated agriculture, and ecological restoration projects (Meng et al., 2020). Meanwhile, although the Loess Plateau is more closely associated with ecological restoration and land management, its semi-arid ecological context also makes it a key research area for analyzing the relationships among afforestation, precipitation constraints, and soil moisture consumption (Jia et al., 2017).

Wetland and river ecosystems primarily show the role of hydrological processes in shaping vegetation patterns and ecological functions. Studies at the watershed scale often focus on the Yellow River and Yangtze River basins, as well as other river systems, to analyze vegetation responses to changes in upstream water inflow, land use, and climate

(Fang et al., 2021). Wetland research more often examines high-latitude wetlands, riparian zones, and lake-surrounding ecosystems, where water-level fluctuation, freeze-thaw cycles, and evapotranspiration processes affect vegetation growth (Zhang et al., 2024). The Amazon, as a large watershed and tropical humid ecosystem, is also used to analyze relationships among precipitation anomalies, drought events, and forest productivity (Yang et al., 2025). These examples explain why water-sensitive ecosystems appear repeatedly in this cluster.

4. Discussion

4.1. Spatial heterogeneity and observational uncertainty

The synthesis of existing research and the supplementary MODIS-era analysis shows that global greening is no longer a localized phenomenon observed in isolated regions. It is a broad signal repeatedly identified in the context of global change. Fig. 9 summarizes screening-level trend classifications for NDVI, LAI and GPP using Sen's slope and the Mann-Kendall test, together with global area-weighted annual means. The analysis was conducted in Google Earth Engine (GEE) using MOD13Q1 NDVI, MOD15A2H LAI and MOD17A3HGF GPP products for 2001-2024. Detailed processing steps are provided in Appendix C.

According to the trend maps generated in this study using Google Earth Engine (Fig. 9), relatively continuous positive greening trends are primarily concentrated in eastern and southern China, India, mid-to-high latitudes of Europe, parts of North Asia, and certain regions of southern Africa and Australia. These areas show a high degree of greening consistency across different remote sensing indicators (Zhang et al., 2021). In particular, China and India have been the most significant contributors to the global net increase in leaf area over the past two decades, accounting for approximately one-quarter and 6.8% of the global increase, respectively. These trends are closely linked to China's afforestation projects

and agricultural intensification, as well as enhanced land management in India (Cao et al., 2025; Chen et al., 2019). Meanwhile, the widespread greening of high-latitude northern regions and the Pan-Arctic tundra is more closely associated with warming and the lengthening of the growing season (Berner et al., 2020). The mapped significance classes should be interpreted as a consistent annual-scale screening result rather than as a phenology-specific or causal attribution analysis.

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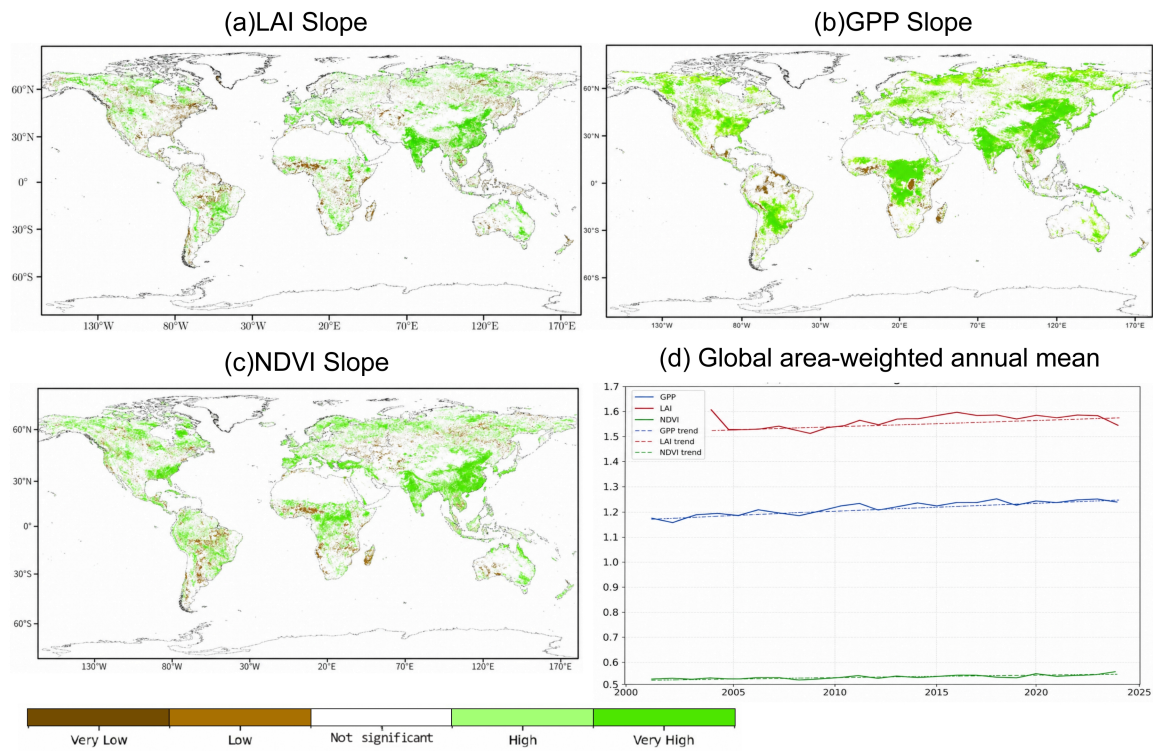
620 [Table 3](#)

621 Common remote sensing sensors and product characteristics used in global greening research.

Sensors	Products	Spatial-Temporal Resolution	Dates	Data Sources
NOAA-AVHRR	AVHRR NDVI time series	~1.1 km; daily	1981—	https://ladsweb.modaps.eosdis.nasa.gov/missions-and-measurements/products/N11_AVH13C1
	GIMMS	14 d, 1/12°	1981—	https://daac.ornl.gov/VEGETATION/guides/Global_Veg_Greenness_GIMMS_3G.html
	GVI	7 d, ≥15 km	1982—	https://www.star.nesdis.noaa.gov/smcd/emb/vci/gvps/gvps_background.php
Landsat4/5-TM; Landsat7-ETM+	GWELDYR / GWELDMO	yearly, 30 m; monthly, 30 m	1984—	https://lpdaac.usgs.gov/products/gweldmo_v031/
SPOT-VGT	VGT-S10	10 d, 1 km	1998—	https://dataspace.copernicus.eu/spot-vegetation
PROBA-V	CGLS V3	10 d, 1 km	1999—	https://dataspace.copernicus.eu/proba-v
Terra/Aqua-MODIS	MOD13Q1 / MYD13Q1	16 d, 250 m	2000—	https://modis.gsfc.nasa.gov/data/dataproduct/mod13.php
	MOD13A1 / MYD13A1	16 d, 500 m	2000—	https://modis.gsfc.nasa.gov/data/dataproduct/mod13.php
	MOD13A2 / MYD13A2	16 d, 1 km	2000—	https://modis.gsfc.nasa.gov/data/dataproduct/mod13.php
	VNP13A1	16 d, 500 m	2011—	https://viirsland.gsfc.nasa.gov/Val/VI_Val.html
	VNP13A2	16 d, 1 km	2011—	https://viirsland.gsfc.nasa.gov/Val/VI_Val.html
VIIRS	VNP13A3	Monthly, 1 km	2011—	https://viirsland.gsfc.nasa.gov/Val/VI_Val.html
	VNP13C1	16-day, 0.05°	2011—	https://viirsland.gsfc.nasa.gov/Val/VI_Val.html
	VNP13C2	Monthly, 0.05°	2011—	https://viirsland.gsfc.nasa.gov/Val/VI_Val.html
PROBA	PROBA-V	10 d, 300 m	2016—	https://proba-v.vgt.vito.be/en
Sentinel-2	S2 L1C	5 d, 10 m	2015—	https://sentiwiki.copernicus.eu/web/s2-mission
Sentinel-3-OLCI	Sentinel 3-OLCI	11 d, 300 m	2020—	https://sentiwiki.copernicus.eu/web/s3-olci-instrument
Sentinel-1	C-band SAR	~6 d	2014—	https://sentinels.copernicus.eu/web/sentinel/missions/sentinel-1
ASTER	ASTER standard	15–90 m	1999—	https://asterweb.jpl.nasa.gov/

Sensors	Products	Spatial-Temporal Resolution	Dates	Data Sources
	products			

623



624

625 **Fig. 9. Spatiotemporal patterns of global vegetation greening based on remote sensing indicators: (a) LAI**
626 **Mann-Kendall/Sen trend class, (b) GPP Mann-Kendall/Sen trend class, (c) NDVI Mann-Kendall/Sen**
627 **trend class, and (d) global area-weighted annual mean of LAI, NDVI and GPP during 2001-2024.**

628 **Fig. 9** also shows that many regions have not followed the broader greening tendency.

629 Some areas show browning or degradation, including the western and southwestern United

630 States, Alaska, parts of southern North American coniferous forests, portions of the Amazon,

631 especially the southeastern Amazon, and localized high-latitude or arid fringe areas ([Verbyla](#)

632 [et al., 2011](#)). Southwestern North America has experienced greenness declines under severe

633 drought ([Foster et al., 2022](#)). Long-term NDVI declines have also been reported in northern

634 Alaskan forests, where drought, fire, insect infestation and permafrost change can offset high-

635 latitude greening ([Beck et al., 2011](#)). The Amazon is not uniformly greening; recent studies

636 report weakening resilience in southeastern areas and in regions repeatedly affected by

637 drought, with growing vulnerability and degradation risk ([Van Lierop et al., 2015](#); [Jin et al.,](#)

2020). Global greening is therefore better described as a broad increase with persistent regional browning than as synchronous improvement across all vegetated regions.

Current evidence for global greening is based mainly on remote-sensing indices and their derivatives (Zeng et al., 2022). Table 4 summarizes the main vegetation indices and products used in this literature. Early AVHRR greenness records, and later NDVI, EVI, and LAI products, describe vegetation status through reflectance or canopy-structure information (Xiao et al., 2016). For example, Myneni et al. used satellite-derived signals of photosynthetic activity to identify vegetation increases in high-latitude regions (Myneni et al., 1997). Zhu et al. estimated the extent and drivers of global greening by comparing multiple LAI products with model simulations (Zhu et al., 2016), and Chen et al. assessed the contributions of China and India using MODIS LAI data (Chen et al., 2019). In this sense, global greening is first a remote-sensing signal of stronger vegetation activity. It should not be treated automatically as evidence of higher ecosystem quality. Although the broad greening signal is well supported, its intensity, extent, and ecological meaning still depend on indicator choice, sensor performance, and regional context.

Table 4

Common vegetation indices and products used in global greening research and their structural characteristics.

Indicator / product	Full name	Frequency in this study	Common data sources	Construction / interpretation
NDVI	Normalized Difference Vegetation Index	122	AVHRR, MODIS, Landsat, Sentinel-2, VIIRS and other optical sensors	Calculated from red and near-infrared reflectance: $NDVI = (NIR - Red) / (NIR + Red)$. It is widely used to represent vegetation greenness but may saturate in dense canopies.
LAI	Leaf Area Index	64	MODIS LAI, GLASS LAI, GLOBMAP LAI, PROBA-V/CGLS LAI, or multispectral inversion products	Represents one-sided leaf area per unit ground area. It reflects canopy structure more directly than greenness indices but depends on retrieval algorithms and assumptions.
EVI	Enhanced Vegetation Index	62	MODIS, Landsat, Sentinel-2 and other optical sensors with blue, red and NIR bands	$EVI = 2.5 \times (NIR - Red) / (NIR + 6 \times Red - 7.5 \times Blue + 1)$. It reduces soil and atmospheric influences and is less prone to saturation than NDVI in dense vegetation.
SIF	Solar-Induced Chlorophyll Fluorescence	24	GOME-2, OCO-2, TROPOMI, GOSAT and other fluorescence-related products	Retrieved from high-spectral-resolution observations of vegetation fluorescence signals. It is more directly related to photosynthetic activity than structural greenness indices.
FVC	Fractional Vegetation Cover	16	MODIS, Landsat, Sentinel-2 and classification- or index-based products	Usually estimated from vegetation indices or spectral mixture methods; a common NDVI-based expression is $FVC = (NDVI - NDVI_{soil}) / (NDVI_{veg} - NDVI_{soil})$.
fAPAR / FPAR	Fraction of Absorbed Photosynthetically Active Radiation	9	MODIS FPAR, GLASS FPAR, Copernicus/CGLS products, radiative transfer model inversion	Represents the fraction of incoming photosynthetically active radiation absorbed by vegetation canopy, linking canopy structure with energy absorption.
VOD	Vegetation Optical Depth	8	Passive microwave products from AMSR-E, AMSR2, SMOS, SMAP, VODCA and related datasets	Retrieved from microwave radiative transfer models. VOD reflects vegetation water content, aboveground biomass, and canopy structure, but it is not an optical greenness index.
NIRv	Near-Infrared Reflectance of Vegetation	5	MODIS, Landsat, Sentinel-2, VIIRS and other red–NIR optical sensors	Commonly calculated as $NIRv = NDVI \times NIR$. It links vegetation greenness with near-infrared reflectance and is often used as a proxy related to photosynthetic capacity or productivity.
EVI2	Two-band Enhanced Vegetation Index	4	MODIS, Landsat, Sentinel-2, VIIRS and other sensors with red and NIR bands	$EVI2 = 2.5 \times (NIR - Red) / (NIR + 2.4 \times Red + 1)$. It is a two-band alternative to EVI when the blue band is unavailable or unreliable.
DVI	Difference Vegetation Index	4	Optical sensors with red and NIR bands, including AVHRR, MODIS, Landsat and Sentinel-2	$DVI = NIR - Red$. It is simple and sensitive to vegetation amount but is more affected by illumination, soil background and sensor calibration than normalized indices.
kNDVI	Kernel Normalized Difference Vegetation Index	3	Multispectral red and NIR reflectance from MODIS, Landsat, Sentinel-2 and similar sensors	A nonlinear transformation of NDVI using a kernel function; a common simplified form is $kNDVI = \tanh(NDVI^2)$. It aims to enhance sensitivity and reduce saturation effects.
TGI	Triangular Greenness Index	3	RGB imagery, UAV imagery, high-resolution cameras, and visible-band satellite data	A visible-band greenness index based on the triangular relationship among red, green and blue reflectance; it is commonly used when only RGB or visible bands are available.
VCI	Vegetation Condition Index	3	Multi-temporal NDVI records from AVHRR, MODIS, VIIRS and similar sensors	$VCI = (NDVI - NDVI_{min}) / (NDVI_{max} - NDVI_{min})$. It describes the relative vegetation condition compared with its historical range and is often used for drought or vegetation stress monitoring.

4.2. Attributing global greening: climate-change and human-activity interactions

Global greening results from the long-term interaction of climatic conditions, ecohydrological processes, soil properties, topographic constraints, and human activities (Piao et al., 2019). The mechanism framework in Fig. 10 shows that solar radiation, temperature, precipitation, and CO₂ concentration together define the broad climatic conditions for vegetation growth.

Surface runoff, subsurface infiltration, soil physicochemical properties, and topographic factors further influence the redistribution of water and nutrients at the surface and in the root zone (Teng et al., 2023). Disturbances such as urbanization, agriculture, afforestation, and natural disasters also alter vegetation types, land-cover patterns, and local ecological processes through land-use change and management practices (Zeng et al., 2018).

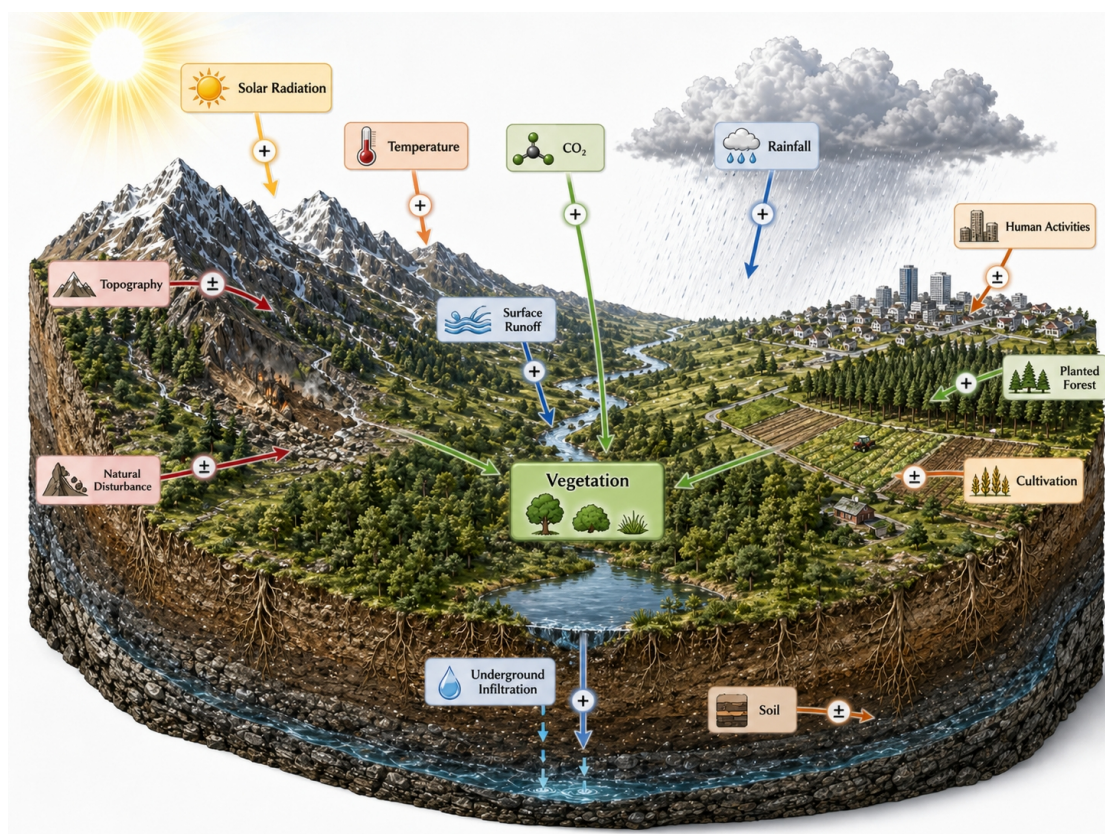


Fig. 10. A comprehensive framework of the driving mechanisms behind global greening: the coupling of climate, land surface processes, and human activities

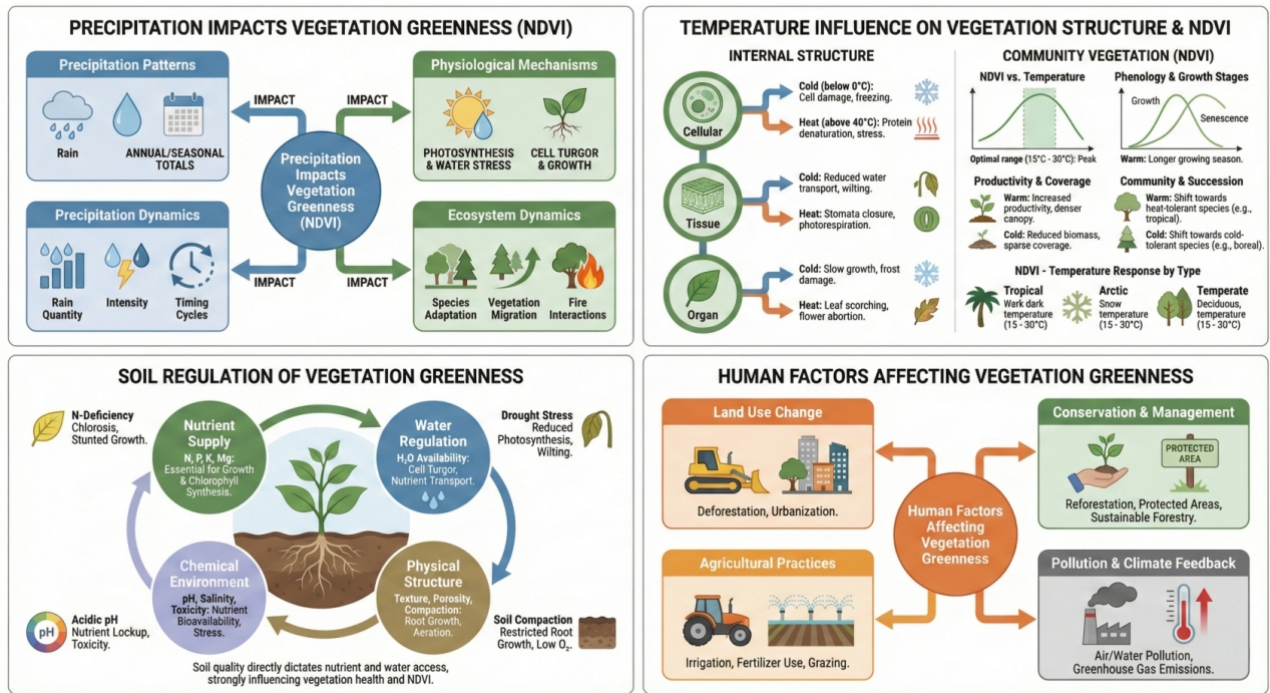


Fig. 11. Diagram of the key mechanisms contributing to global greening: pathways involving precipitation, temperature, soil, and human activities

This causal relationship can be divided into several connected mechanism modules (Fig. 11). First, precipitation influences greenness through its amount, intensity, timing and interannual distribution. It determines soil-water supply and water availability, and it can also alter fire disturbance, species regeneration and growing-season rhythm (Yin et al., 2025). Second, temperature has a dual effect. Warming within a suitable range can promote photosynthesis, extend growing seasons and increase vegetation cover, whereas warming beyond physiological or moisture thresholds may increase transpiration, heat stress and growth limitation (Chu et al., 2018). Third, soil mediates the translation of climate signals into vegetation responses through nutrient supply, water regulation and physicochemical structure (Li et al., 2023). Finally, human activities reshape ecological processes through land-use change, agricultural management, conservation and restoration, pollution and climate feedbacks, which can either enhance or weaken global greening (Khan et al., 2025).

This decomposition suggests that the attribution of global greening should be understood as the combined effect of interacting mechanism modules rather than as a simple binary contrast between climate-driven and human-driven change (Bueso et al., 2023).

4.3. Future trends and research priorities

Global scenario studies generally project continued greening over the coming decades, but its magnitude, persistence and ecological meaning are less stable and less spatially uniform than once assumed. Future projections indicate that vegetation greenness may increase under different climate scenarios through the 21st century. The increase may also be smaller than conventional multi-model ensemble estimates. Zhao et al. (2020) show that future Earth greening remains a general tendency, but reliability ensemble averaging produces a lower magnitude than conventional multi-model means. This comparison points to possible overestimation in earlier projections, especially when model uncertainty and inter-model reliability are considered more carefully.

Future greening also cannot be equated with a continuous increase in terrestrial carbon sequestration. Bian and Xia (2024) show that the positive relationship between greenness and carbon storage may reach turning points under different emission pathways. Under the three examined emission scenarios, this greenness-carbon relationship is projected to weaken after approximately 2034, 2038, and 2048, respectively. Under the high-emission SSP5-8.5 scenario, vegetation greenness may continue to increase, yet carbon storage efficiency is expected to decline. As much as 71% of vegetated areas may experience decoupling between greenness and carbon storage. This challenges the assumption that a greener land surface necessarily implies a stronger carbon sink. Future global greening research therefore needs to

examine whether increases in vegetation indices are accompanied by improvements in productivity, carbon sequestration, water-use efficiency, and ecological stability.

The water-cycle consequences of vegetation greening may also become more apparent. [Lu et al. \(2021\)](#) suggest that terrestrial evapotranspiration in China will increase during 2020-2099 under all climate change scenarios. As radiative forcing intensifies, however, the sensitivity of land evapotranspiration to vegetation cover is expected to decrease. Vegetation greening may continue to influence regional water and energy exchanges, but the strength and efficiency of this influence may weaken under stronger warming scenarios. This result fits the mechanism discussed in this study: greening is not only a carbon-cycle issue but also a water-energy coupling process. In regions where atmospheric evaporative demand, vapor pressure deficit, and water stress increase, additional vegetation growth may no longer produce proportional ecological benefits and may even intensify water consumption pressure.

Regional scenario studies point to strong spatial divergence in future greening. [Bathiany et al. \(2014\)](#) indicate that rising atmospheric CO₂ concentrations may promote vegetation cover expansion in the Sahara and surrounding regions before 2100. The effect may not persist indefinitely. On longer timescales, especially from 2100 to 2300, CO₂ fertilization may weaken, and vegetation cover may decline again, particularly in the western Sahara. CO₂-induced greening may therefore be transient in some dryland ecosystems. [X. Zhang et al. \(2024\)](#) suggest that vegetation productivity in arid regions worldwide may continue to increase in the coming decades, mainly because of CO₂ fertilization, and that less than 4% of dryland areas are projected to undergo desertification despite increasing aridity. These results point to dryland responses that are more complex than a simple warming-drying-degradation pathway. Still, they should not be read as evidence that dryland ecosystems are free from risk, because enhanced productivity may coexist with increasing water stress, reduced resilience, and higher vulnerability to extreme drought.

Some regional forest ecosystems may face clear browning or degradation risks under future climate change. [Daghestani et al. \(2024\)](#) project that, by 2050, climate-change-induced drought may substantially reduce forest greenness in the Zagros oak forests of western Iran, with declines of approximately 25% under RCP8.5 and 15% under RCP2.6. Even moderate climate scenarios may negatively affect drought-sensitive forests, while high-emission pathways may further amplify vegetation decline. [B. Zheng and Yu \(2024\)](#) show that, under SSP119, SSP245, and SSP585, forest NDVI on Hainan Island is projected to show an overall downward trend. These projections point to humid tropical forest ecosystems that may not benefit from future warming or CO₂ enrichment; instead, they may become increasingly constrained by hydrothermal shifts, extreme climate events, or declining climatic suitability.

Scenario evidence changes the central question for future global greening research. The issue is no longer only whether the Earth will continue to become greener, but where greening will continue, where it will stagnate or reverse, and whether increased greenness will still correspond to improved ecosystem function. At the global scale, greening may persist as an overall trend, but its rate may be lower than previously expected and its functional benefits may weaken. At the regional scale, future trajectories may diverge sharply. Some drylands and CO₂-sensitive regions may continue to show increases in vegetation productivity, whereas drought-sensitive forests, tropical ecosystems and regions facing stronger water stress may experience greenness decline or functional degradation. Future global greening is therefore more likely to involve broad continuation with regional divergence and functional decoupling than uniform ecological improvement.

Several research directions follow from this changing understanding. Greenness indicators need to be examined together with carbon storage, productivity, evapotranspiration, soil moisture, vegetation optical depth, and solar-induced chlorophyll fluorescence to distinguish structural greening from functional improvement. Earth system models, remote

sensing observations, and machine-learning approaches can be linked more explicitly to evaluate nonlinear vegetation responses to CO₂ fertilization, warming, drought, and human land management. Ecological thresholds and turning points also deserve closer attention, especially where greening may decouple from carbon sequestration or where water stress may offset the benefits of increased vegetation cover. Regional comparisons across drylands, tropical forests, high-latitude ecosystems, agricultural regions, and ecological restoration zones will be especially useful. Linking future greenness changes with carbon, water, and ecosystem resilience can move global greening research from trend projection toward a more grounded assessment of ecological sustainability under climate change.

5. Conclusions

This study used a bibliometric review, supported by supplementary evidence synthesis, to trace how global greening research has developed, what themes now organize the field, and how its methods have changed. The bibliometric results show steady growth since the mid-to-late 1980s and rapid expansion after 2010. The field has moved from early phenomenon detection to questions about mechanisms, ecological consequences, and future trajectories. China and the United States account for the largest publication output, although their roles in collaboration networks still need to be evaluated with co-authorship metrics such as collaboration intensity, total link strength, and network centrality. The Chinese Academy of Sciences, Beijing Normal University, the University of Chinese Academy of Sciences, Peking University, NASA, and Boston University are major contributors at the institutional level. These patterns should be treated as publication-based evidence, not definitive measures of collaboration strength. Disciplinary and journal distributions show that global greening research is rooted in remote sensing, environmental science, ecology, and climate-change research.

Keyword clustering, co-citation analysis, and thematic evolution point to three main directions. One concerns climate change, CO₂ fertilization, water-energy constraints, and carbon-cycle feedbacks. Another centers on methodological support from multi-source remote sensing, long-term vegetation indices, Google Earth Engine, machine learning, and ecological models. A third focuses on spatiotemporal responses linked to phenology, ecological restoration, agricultural management, land degradation, and regional heterogeneity. The field has moved beyond asking whether vegetation is becoming greener. It now asks where greening occurs, why it occurs, whether it reflects functional ecosystem improvement, and whether it can last under future climate and land-use change.

The supplementary MODIS-era trend analysis supports the literature-based conclusion that global greening is spatially uneven and sensitive to method choices. NDVI, LAI and GPP trends show widespread but uneven vegetation change, and their agreement varies by region and indicator. These analyses are illustrative, not formal attribution tests. Indicator choice, sensor characteristics, spatial resolution, temporal coverage, masking strategy and trend-detection method all shape greening estimates. NDVI and LAI should therefore not be equated directly with ecosystem quality, carbon sequestration or ecological resilience. Stronger conclusions about functional consequences require long-term optical records, functional indicators, process-based models, ground observations, field experiments and independent datasets.

Several limitations remain. First, the bibliometric results depend on the search strategy, database coverage, language restriction and keyword standardization. Relevant studies outside the selected Web of Science Core Collection records may have been missed, and the search strategy was not tested through a formal sensitivity analysis. Second, country and institutional patterns are based mainly on publication and affiliation information, so they

cannot fully describe collaboration networks. Third, the supplementary remote-sensing analyses provide context for the review rather than a full global attribution framework. The Mann-Kendall/Sen results in Fig. 9 use annual calendar-year composites, a common 5-km export scale, an ecological mask and screening-level significance thresholds, but they do not resolve growing-season phenology, serial autocorrelation, field validation or causal mechanisms. Further work needs to connect long-term optical records with functional indicators such as SIF, VOD, NIRv, carbon storage, evapotranspiration and soil moisture; combine remote sensing with ground observations, ecological experiments and Earth system models; and strengthen regional comparisons across drylands, tropical forests, high-latitude ecosystems, agricultural regions and ecological restoration zones. Such integration would help global greening research move from greenness characterization toward stronger assessments of ecosystem function, resilience and sustainability under global change.

CRediT authorship contribution statement

Declaration of competing interest

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Data availability

Bibliometric records were retrieved from the Web of Science Core Collection using the search strategy described in Appendix A. The remote-sensing products used for the supplementary illustrations are publicly available through Google Earth Engine and the data sources listed in Tables 3 and 4. The Google Earth Engine script used to generate Fig. 9, together with the exported 0-4 trend-class maps and annual area-weighted mean table, should be provided as supplementary code and data at submission or made available by the authors on reasonable request.

Appendix A. Bibliometric data collection

This study used the Web of Science Core Collection (WOSCC) as the primary source of bibliographic data. To ensure academic representativeness, traceability and comparability, the search was conducted in citation-indexed WOSCC databases, including SCI-EXPANDED, SSCI and ESCI. The search was conducted on 15 May 2025. Because global greening research is highly interdisciplinary and involves remote-sensing monitoring, vegetation dynamics, climate change, ecological feedbacks and human activities, the retrieval strategy was designed to cover both greening-related and remote-sensing-related terms. The strategy was intentionally broad, but it should not be read as an exhaustive census of all vegetation-change literature because database coverage, language restriction and query wording can influence the retained record set.

The search query was formulated as follows:

TS=("global greening" OR "Earth greening" OR "greening Earth" OR "vegetation greening" OR "vegetation greenness" OR "vegetation dynamics" OR "vegetation activity" OR "NDVI trend*" OR "LAI trend*" OR "terrestrial greening") AND TS=("remote sensing" OR "remote sensing ecological index")

The document types were restricted to Article and Review, and the language was restricted to English. The publication period covered 1980-2025. The initial search returned 704 records. After removing 206 duplicate records, 498 records remained. Titles, abstracts, and keywords were then checked to confirm that the retained records were relevant to remote-sensing-based global greening. Records outside this scope, such as studies focused only on engineering applications of remote sensing, local image processing without vegetation-change interpretation, or phenological identification without relevance to vegetation greening, were not included in the retained set. The final sample contained 498 records for bibliometric analysis, including 351 articles and 147 reviews.

The final literature sample was used to analyze publication trends, country and institutional output, disciplinary and journal distributions, keyword evolution, co-citation structures, and method frequencies. Country and institutional statistics were calculated from author-affiliation information in WOSCC. Unless stated otherwise, publication counts, keyword frequencies, affiliation contributions, and method frequencies used full counting: each occurrence or affiliation was counted once when it appeared in a record. Because affiliation-based statistics do not fully measure collaboration, these results should be read as patterns of publication output and affiliation distribution, not as definitive evidence of collaboration strength.

Appendix B. Literature analysis methods

To map the knowledge structure, thematic organization, and evolution of global greening research, this study combined bibliometric analysis, knowledge-graph analysis, keyword co-occurrence analysis, co-citation analysis, and manual thematic synthesis. The workflow had four steps: data retrieval and screening, metadata cleaning and standardization, quantitative bibliometric analysis, and interpretation of the visualized networks.

First, the bibliographic records exported from WOSCC were standardized before analysis. Duplicate records were removed, and author keywords, Keywords Plus, institutional names, country names and journal names were checked for consistency. Term cleaning used conservative normalization rules: singular and plural forms were merged, obvious spelling or capitalization variants were standardized, abbreviations were linked to full terms where the meaning was unambiguous, and near-synonymous greening terms such as “vegetation greening” and “greening vegetation” were grouped. Obvious irrelevant terms and overly general words were excluded from keyword interpretation to improve thematic clarity. Ambiguous terms were retained rather than forced into a cluster when their meaning could not be resolved from titles, abstracts or keywords.

Second, publication trends were examined by counting records by year. Country and institutional distributions were calculated from author affiliations. Journal and disciplinary distributions were analyzed using WOSCC categories and journal metadata. Full counting was used for publication counts, keyword frequencies, affiliation contributions and method frequencies because the purpose was to map the presence and distribution of research activity, not to allocate fractional credit among authors or institutions. These results were used to identify the main contributors, disciplinary bases and publication outlets in global greening research.

Third, keyword co-occurrence and thematic clustering were used to identify major research topics and their changes over time. High-frequency keywords were grouped through bibliometric visualization and manual interpretation. The synthesis was organized around remote-sensing observations and data sources, climate and ecological feedbacks, regional case studies and human management, and water constraints and ecological vulnerability. Temporal keyword analysis was then used to trace the field's movement from vegetation-index monitoring and greenness detection toward mechanism analysis, ecological feedback assessment, machine learning, high-resolution remote sensing and future scenario projection. The clusters were interpreted as descriptive research themes, not as statistically independent communities.

Fourth, co-citation analysis was used to identify foundational publications and knowledge bases. Highly co-cited references were treated as key nodes in methodological development, global greening detection, driver attribution, and ecological feedback analysis. The co-citation network was then used to describe the field's intellectual structure and its shift from remote-sensing monitoring toward mechanism- and assessment-oriented studies.

Visualization and network analyses were conducted using VOSviewer (<https://www.vosviewer.com>) and CiteSpace (<https://cluster.cis.drexel.edu/~cchen/citespace/download/>). VOSviewer was mainly used for co-occurrence and co-citation network visualization. Network outputs were treated as descriptive maps of the reviewed literature rather than direct evidence of ecological processes, collaboration quality or scientific importance. Cluster appearance can be affected by threshold settings, thesaurus choices, counting rules and layout algorithms; therefore, broad patterns were interpreted only when they were consistent with keyword frequencies, co-citation context and manual reading of representative records. Country and institutional statistics based on author affiliations do not necessarily represent actual collaboration intensity.

The bibliometric results were therefore interpreted together with literature-based synthesis and supplementary remote-sensing illustrations to support a cautious reading of research trends, methodological uncertainties and future directions in global greening studies.

Appendix C. Supplementary remote-sensing analysis

To support the discussion of spatial heterogeneity and observational uncertainty, this study added a MODIS-era remote-sensing analysis in Google Earth Engine. Three MODIS Collection 6.1 products were used for 2001-2024: MOD13Q1 NDVI, MOD15A2H LAI and MOD17A3HGF GPP. MOD13Q1 NDVI has a native resolution of 250 m and a 16-day interval; the NDVI band was multiplied by 0.0001 and filtered using DetailedQA bits 0-1 equal to 0. MOD15A2H provides Terra LAI/FPAR observations at 500 m and 8-day intervals; the Lai_500m band was multiplied by 0.1 and filtered using FparLai_QC bits 0-1 equal to 0. MOD17A3HGF provides annual GPP at about 500 m; the Gpp band was multiplied by 0.0001 and invalid negative pixels were removed.

Because these products have different native spatial resolutions, all maps, diagnostic statistics and global area-weighted annual means were produced at a common 5-km analysis/export scale in Google Earth Engine. The 5-km scale was applied through GEE reduction and export operations, rather than by building an independent fishnet grid. No nearest-neighbor or mean resampling rule was imposed manually; GEE handled scale-based raster aggregation or resampling during reduction and export. The 5-km outputs should therefore be read as coarse illustrative products, not fine-resolution trend maps.

Quality control and ecological masking were applied before annual aggregation. The ecological mask used MCD12Q1 land/water and land-cover information from 2015. Water pixels were removed using LW, and LC_Type1 classes 15 (snow and ice) and 16 (barren or sparsely vegetated land) were excluded. A long-term MOD13Q1 NDVI climatology for 2010-2020 was also used to remove extremely low-vegetation surfaces with mean NDVI below 0.05. Croplands were retained because agricultural management is an important part of observed greening signals. Annual NDVI and LAI values were calculated as calendar-year arithmetic means of all quality-controlled observations, rather than maximum-value composites, growing-season means or seasonal integrals. For GPP, the annual MOD17A3HGF Gpp band was used directly after scaling.

Global annual means were calculated as area-weighted averages with `ee.Image.pixelArea()`, so each valid pixel contributed according to its surface area and latitude-related pixel-area differences were accounted for implicitly. Pixel-level trends were estimated from annual time series using Sen's slope and tested with the Mann-Kendall test. Significance was classified with Z thresholds of 1.96 and 2.58, corresponding to 95% and 99% confidence levels. Exported trend classes were coded as 0 for non-significant or masked areas, 1 for significant decrease, 2 for highly significant decrease, 3 for significant increase and 4 for highly significant increase. The annual mean table and 0-4 trend-class maps were exported from GEE for NDVI, LAI and GPP.

Several methodological boundaries follow from this design. The trend analysis used annual calendar-year composites and did not separate growing seasons between the Northern and Southern Hemispheres. The Mann-Kendall implementation used the classical variance form and did not explicitly correct for serial autocorrelation or tied ranks. Fig. 9 should therefore be read as an illustrative MODIS-era screening analysis of annual-scale greenness and productivity tendencies, not as a phenology-specific assessment or causal attribution test.

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